

Paragliding Intelligence Platform

Volume 01 - Technical Product Requirements Document v2

A paragliding-first meteo, risk, XC planning, digital-twin, and pilot decision platform designed to exceed general weather apps.

Document status: Rebuilt technical PRD, replacing outline/padding versions.

Audience: founder, product owner, engineering lead, UX lead, weather/GIS engineer, ML engineer, mobile engineer, investor technical reviewer.

Scope: This document defines the product, technical product requirements, feature contracts, acceptance criteria, tier entitlements, data dependencies, and implementation backlog. System design, database schema, API OpenAPI, weather engine internals, ML details, infrastructure, security, and QA will be expanded in later volumes.

Attribute	Value
Version	v2 Technical PRD
Prepared for	Paragliding meteo platform project
Date	2026-06-04
Primary product promise	Tell pilots where, when, and whether to fly - not just what the weather is.
Core differentiator	Site-specific decision intelligence using weather models, terrain, pilot skill, live observations, launch digital twins, and safety scoring.

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1. Executive Summary and Product Thesis

The product is a paragliding-specific weather and decision platform. It starts with a Windy/Ventusky-class interactive meteo map, but the strategic goal is not to duplicate general weather visualisation. The strategic goal is to become the pilot decision layer: a system that combines forecast models, terrain, launch geometry, airspace, pilot skill, live observations, digital twin corrections, and safety rules to produce actionable recommendations.

The product shall answer four operational questions better than general weather apps:

- Should I drive to this launch today, or is another launch safer and more productive?
- What is the best launch window and how does it change by pilot level?
- What can go wrong: gusts, rotor, valley wind, Foehn, overdevelopment, cloudbase, airspace, retrieve risk?
- What is the expected XC potential: thermal quality, drift, cloud streets, route choices, and safe landing options?

The product is not marketed as absolute safety certification. It is a decision-support and risk-intelligence tool. Every recommendation must expose reasons, forecast confidence, source age, and uncertainty.

1.1 Product Positioning

Dimension	General weather apps	Paragliding Intelligence Platform requirement
Primary output	Weather map layer or point forecast	GO/MAYBE/NO-GO decision with reasons, confidence, and skill-level adjustment
Weather model usage	Display raw or lightly processed model output	Fuse models, correct locally, compare forecast vs reality, and learn site bias
Terrain role	Map background and altitude layer	Core feature: slope, aspect, ridge orientation, valley wind, rotor zones, glide terrain
User profile	Mostly generic user settings	Pilot skill, wing class, risk tolerance, retrieve mode, live tracking preferences
Monetization basis	Forecast layers and alerts	Decision intelligence, digital twins, school/club/event operations, API, hardware data network

1.2 Product Principle: Data Must Become Decisions

Every layer must either improve a decision, reduce risk, increase forecast confidence, or create explainability. A layer that looks attractive but does not change pilot decisions should be lower priority than a less visible feature that prevents bad launch choices.

Decision pipeline:

Forecast data + terrain + launch metadata + pilot profile + live reality + history

- > normalized feature vector
- > site-specific correction
- > risk classifier
- > flyability / XC / safety scores
- > GO/MAYBE/NO-GO recommendation with explanation

2. Competitive Baseline and Opportunity

The market already contains excellent visual weather and flying apps. The gap is not missing wind animation. The gap is integrated, paragliding-specific decision intelligence that combines launch compatibility, terrain, live reality, skill level, airspace, and historical correction.

Competitor / reference	Observed capability	Opportunity for this platform
Windy / Windy API	Layered weather visualisation, particles, isolines, programmatic map layers, point	Do not compete only on visual layers. Build paragliding-specific scoring, site

	forecasts	correction, and GO/NO-GO workflow on top of comparable map quality.
Ventusky	Large number of weather layers, live radar/lightning, webcams, multi-model weather presentation	Compete by adding launch-specific decision outputs, pilot personalization, digital twins, and XC route planning.
XCTrack	Flight computer, thermal assistant, terrain maps, airspace support, livetracking, competition navigation	Complement/compete with planning-before-launch and cloud decision intelligence, not only in-flight instrumentation.
Paragliding Map	Large launch/landing/site database, weather stations, webcams, valley wind directions, forecasts	Compete by converting site data into scoring, route, risk, and AI decisions rather than only discovery.
XC Therm / AeroXC style products	Thermal and XC-oriented weather maps with altitude layers and forecasts	Compete via broader app ecosystem, digital twins, school/club/event modules, and ML correction.

2.1 Differentiation Requirements

- The app shall never present a single score without a reason list and confidence indicator.
- The app shall show forecast vs observed reality whenever station, pilot report, webcam, or track data exists.
- The app shall learn persistent model bias per launch and altitude band.
- The app shall provide separate Flyability, Safety, and XC Potential scores instead of collapsing all meaning into one number.
- The app shall rank sites by user drive radius, pilot level, launch compatibility, and risk, not just nearby weather.
- The app shall include post-flight debrief loops so IGC/GPX tracks improve both pilot coaching and local forecast correction.

2.2 Moat Hypothesis

The defensible asset is not the global weather data. The defensible asset is the corrected, site-level, paragliding-specific outcome database: forecasts, actual launch reports, station readings, webcam-derived conditions, IGC flight outcomes, incident markers, and pilot feedback. Over time this becomes a launch digital twin that a generic weather app does not have.

3. North-Star Product Requirements

Requirement ID	Requirement	Priority	Owner service / domain	Acceptance signal
NS-001	Show an animated global/regional weather map with wind, gust, rain, cloudbase, thermals, CAPE/storm, visibility, and altitude selector.	P0	Map Client + Weather API	Pilot can scrub 72h timeline and switch altitude in <1s after cache warm.
NS-002	Compute hourly site-specific Flyability, Safety, and XC Potential for every supported launch.	P0	Decision Engine	Each score includes reasons, confidence, source model age, and pilot-level adjustment.
NS-003	Rank flyable sites within a chosen travel radius and time window.	P0	Recommendation Engine	User can ask: "Where can I fly within 2h drive tomorrow?" and receive ranked results.
NS-004	Use live reality signals: stations, pilot reports,	P1	Reality Layer	Site page shows forecast-vs-reality delta

	webcams, radar/lightning.			and last updated timestamps.
NS-005	Support site digital twins that learn forecast bias and local microclimate.	P1	Digital Twin + ML	For sites with enough data, display corrected forecast and explain bias.
NS-006	Provide AI briefing grounded only in structured backend data.	P1	AI Briefing Service	AI response must cite exact data fields used and must not invent weather.
NS-007	Enable paid tiers with entitlement enforcement across API, UI, alerts, and model access.	P0	Billing + Entitlement	Pro features are blocked by server-side entitlement, not only UI gating.
NS-008	Support in-flight safety alerts and tracking integrations.	P2	Tracking + Alerting	Pilot receives warnings for airspace proximity, worsening weather, or emergency signals.

3.1 Product Safety Boundaries

- Recommendations must be phrased as decision support, not official clearance to fly.
- A NO-GO recommendation must never be hidden behind a paid tier.
- If data is stale, missing, or conflicting, the system must lower confidence and display why.
- Beginner mode shall be conservative by default and cannot be silently loosened by marketing experiments.
- AI may summarize and explain but shall not override deterministic safety gates without explicit uncertainty language.

4. Personas and Safety Modes

Persona	Need	Default safety mode	Core paid value
Beginner EN-A pilot	Avoid bad launch choices and understand basic conditions	Very conservative	Beginner-safe site ranking, launch window, instructor- style explanation
Intermediate weekend pilot	Find good sites without wasting a drive	Conservative/standard	Advanced alerts, unlimited favorites, site recommendation
XC pilot	Plan routes and understand thermal quality	Standard/adaptive	XC potential, route weather, thermal chains, cloudbase, drift
Competition pilot	Task planning and risk visibility	Adaptive/expert	Task optimizer, competitor weather dashboard, route risk overlays
Instructor / school	Decide where students can safely train	School policy mode	Student skill matching, daily briefing, attendance/weather records
Club safety officer	Coordinate members and local sites	Club policy mode	Shared observations, incidents, site maintenance and safety alerts
Rescue / retrieve team	Know pilot location and likely drift/landing area	Operational mode	Live tracking, last known position, wind reconstruction, search area estimate

4.1 Safety Mode Parameters

Parameter	Beginner	Intermediate	Advanced/XC	School configurable
Max launch wind	12-18 km/h default depending site	18-24 km/h	Site and wing dependent	Instructor-defined per student level
Max gust spread	<=6 km/h	<=10 km/h	<=14 km/h with warnings	Policy-defined
Crosswind tolerance	Low	Medium	Higher if launch wide and site supports it	Student-specific
Thermal strength caution	Strong thermals reduce beginner score	Moderate thermals accepted	Strong thermals may improve XC score	Training phase dependent
Cloudbase caution	Low cloudbase reduces score	Low cloudbase warning	XC planning optimization	Instructor-defined

5. Feature Catalogue with Technical Contracts

Each feature below defines the product behavior and the technical contract that later architecture, database, and API volumes must implement. IDs are stable and should be used in tickets, roadmaps, tests, and API mapping.

F-001 - Windy/Ventosky-class Weather Map

Field	Technical requirement
Priority	P0
Owner service(s)	map-client, weather-tile-service, forecast-api
Product behavior	Interactive 2D/3D map with timeline, altitude selector, wind particles, vector layers, raster overlays, launch icons, and site status coloring.
Inputs	forecast_cube, tile_cache, launch_status, user_entitlements
Outputs	map viewport, layer legend, time cursor, site markers, value picker
Primary APIs	GET /v1/map/layers; GET /v1/tiles/{layer}/{z}/{x}/{y}; GET /v1/forecast/picker?lat=&lon=&time=&level=
Acceptance criteria	72h timeline works on MVP; Pro supports 10-14 days; altitude layers render without full page reload; each map value displays timestamp and source model.

F-002 - Launch and Landing Site Intelligence

Field	Technical requirement
Priority	P0
Owner service(s)	site-service, gis-service
Product behavior	Global site registry with launch/landing points, wind directions, obstacles, rotor notes, access rules, skill requirements, local club ownership, webcams, and weather station links.
Inputs	manual curation, imports from open/collaborative databases, club admin edits, terrain analysis
Outputs	site detail page, launch compatibility engine, ranked search results
Primary APIs	GET /v1/sites; GET /v1/sites/{id}; POST /v1/sites/{id}/report; PATCH /v1/sites/{id} (moderated)
Acceptance criteria	Site must store geometry, altitude, aspect, allowed wind sector, min/max wind, max gust, difficulty, hazards, source quality, last

	verified date.
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F-003 - Flyability Score

Field	Technical requirement
Priority	P0
Owner service(s)	decision-engine
Product behavior	Hourly score indicating whether a launch is practically usable for a selected pilot mode. Separate from risk and XC potential.
Inputs	wind speed/direction/gust, launch aspect, precipitation, cloudbase, visibility, thermal intensity, pilot mode, site limits
Outputs	score 0-100, status GO/MAYBE/NO-GO, reason list, confidence
Primary APIs	GET /v1/flyability/site/{site_id}?from=&to=&pilot_mode=
Acceptance criteria	Every score includes top positive and negative factors; score changes when pilot mode changes; stale data lowers confidence.

F-004 - Safety Risk Score

Field	Technical requirement
Priority	P0
Owner service(s)	risk-engine
Product behavior	Dedicated risk score for conditions that may be flyable but unsafe: gusts, rotor, Foehn, valley wind, wave, thunderstorms, overdevelopment, airspace, low visibility.
Inputs	forecast cube, terrain exposure, pressure gradient, CAPE/shear, radar/lightning, valley geometry, station deltas
Outputs	risk 0-100, safety flags, hard blockers, warnings
Primary APIs	GET /v1/risk/site/{site_id}; GET /v1/risk/route
Acceptance criteria	Hard blockers override high flyability; risk reasons are structured so alerts and AI can consume them.

F-005 - XC Potential Score

Field	Technical requirement
Priority	P1
Owner service(s)	xc-engine, thermal-engine
Product behavior	Estimates cross-country potential by route corridor, expected climbs, cloudbase, drift, wind at altitude, trigger likelihood, and landing density.
Inputs	thermal grid, cloudbase grid, wind profile, terrain triggers, no-land zones, retrieve constraints
Outputs	XC score, suggested launch window, route suggestions, expected distance class
Primary APIs	GET /v1/xc/potential/site/{site_id}; POST /v1/xc/route-plan
Acceptance criteria	Can rank a 20/50/100 km day; returns not enough data when thermal inputs unavailable instead of hallucinating.

F-006 - Thermal Trigger Map

Field	Technical requirement
Priority	P1
Owner service(s)	thermal-engine, terrain-service

Product behavior	Map of likely thermal trigger zones using sun angle, slope/aspect, landcover, cloud cover, lapse rate proxies, and historical climb traces.
Inputs	DEM, slope/aspect, solar radiation, cloud cover, temperature/dewpoint, IGC climbs, landcover
Outputs	thermal strength layer, trigger confidence, climb band by time
Primary APIs	GET /v1/thermals/tiles; GET /v1/thermals/profile?lat=&lon=
Acceptance criteria	Layer separates raw model thermal potential from learned trigger history; confidence shown per region.

F-007 - Cloudbase and Ceiling Planner

Field	Technical requirement
Priority	P0
Owner service(s)	forecast-processing
Product behavior	Computes site and route cloudbase/ceiling relevance; warns about low cloudbase, overdevelopment, fog, and terrain obscuration.
Inputs	temperature, dewpoint, cloud layers, elevation, humidity profile, satellite/cloud observations
Outputs	cloudbase estimate, clearance above launch, route ceiling profile
Primary APIs	GET /v1/cloudbase/site/{site_id}; GET /v1/cloudbase/route
Acceptance criteria	Displays cloudbase relative to launch and landing altitude, not only absolute MSL.

F-008 - Live Reality Layer

Field	Technical requirement
Priority	P1
Owner service(s)	reality-service
Product behavior	Merges station observations, pilot reports, webcam analysis, radar, lightning, and live tracks into a now layer that validates or contradicts forecasts.
Inputs	weather stations, app reports, webcams, radar/lightning, live tracking telemetry
Outputs	forecast-vs-reality delta, confidence adjustment, warnings
Primary APIs	GET /v1/reality/site/{site_id}; POST /v1/reports/conditions
Acceptance criteria	Each live signal has age, source type, trust score, and effect on decision confidence.

F-009 - Launch Webcam AI

Field	Technical requirement
Priority	P2
Owner service(s)	webcam-service, cv-service
Product behavior	Ingests webcams and detects visibility, cloud cover, launch obscuration, precipitation signs, windsock movement where visible, and pilots on launch.
Inputs	webcam stills/video, site mask, horizon mask, manual validation
Outputs	visibility estimate, cloud growth flag, launch visible flag, image evidence
Primary APIs	GET /v1/webcams/site/{site_id}; GET /v1/webcams/{id}/analysis

Acceptance criteria	AI result must expose confidence; low confidence cannot drive hard NO-GO alone without corroborating data.

F-010 - Site Digital Twin

Field	Technical requirement
Priority	P1
Owner service(s)	digital-twin-service, ml-platform
Product behavior	Self-learning model per site storing forecast error, station reality, track outcomes, pilot reports, incidents, and seasonal local patterns.
Inputs	historical forecast cubes, station obs, reports, IGC/GPX outcomes, site metadata, incident flags
Outputs	corrected forecast, local bias, confidence, microclimate notes
Primary APIs	GET /v1/digital-twins/site/{site_id}; GET /v1/forecast/corrected/site/{site_id}
Acceptance criteria	Only enabled after minimum sample size; displays training data count and validation accuracy.

F-011 - AI Flight Briefing

Field	Technical requirement
Priority	P1
Owner service(s)	ai-briefing-service, context-builder
Product behavior	Natural-language briefing that explains best window, risks, model disagreement, confidence, and alternative sites. The LLM only uses structured backend facts.
Inputs	site forecast, scores, risk flags, pilot profile, favorites, drive radius, live reality
Outputs	briefing text, structured recommendation, citations to fields
Primary APIs	POST /v1/ai/briefing
Acceptance criteria	Prompt must include a system rule: never invent weather; output includes data freshness and uncertainty; deterministic safety gates are passed separately.

F-012 - AI Site Discovery

Field	Technical requirement
Priority	P1
Owner service(s)	recommendation-engine, ai-briefing-service
Product behavior	User asks where to fly near a city or within a drive radius; system ranks sites by conditions, safety, travel time, skill match, and confidence.
Inputs	user location, travel radius/time, site scores, road travel estimates, pilot profile
Outputs	ranked sites, reasons, map route, alternative windows
Primary APIs	POST /v1/recommendations/sites
Acceptance criteria	Shows why a lower-weather-score site may rank higher due to safety or launch compatibility.

F-013 - AI Post-flight Debrief

Field	Technical requirement
Priority	P2
Owner service(s)	debrief-service, flight-analysis-service
Product behavior	Analyzes IGC/GPX track, climbs, glide efficiency, decisions vs forecast, landing choices, route alternatives, and coaching points.
Inputs	IGC/GPX, forecast at flight time, terrain, thermals, pilot goal
Outputs	flight report, mistakes/opportunities, replay annotations
Primary APIs	POST /v1/flights/upload; GET /v1/flights/{id}/debrief
Acceptance criteria	Debrief distinguishes factual track metrics from coaching inference; supports private-by-default storage.

F-014 - Airspace Guardian

Field	Technical requirement
Priority	P1
Owner service(s)	airspace-service, tracking-service
Product behavior	Airspace map, pre-flight warnings, in-flight proximity alerts, altitude conflicts, temporary activations, and country-specific data sources.
Inputs	OpenAIP/national airspace data, NOTAM-like data where available, GPS track, barometric/GNSS altitude
Outputs	airspace polygons, warnings, voice alerts
Primary APIs	GET /v1/airspace/tiles; GET /v1/airspace/conflicts; WS /v1/tracking/live
Acceptance criteria	Warns pre-flight and live; airspace data age visible; offline cache required for in-flight use.

F-015 - Live Tracking and Retrieve

Field	Technical requirement
Priority	P2
Owner service(s)	tracking-service, websocket-gateway
Product behavior	Live pilot position, retrieve team view, emergency share link, optional public/private modes, FANET/OGN/other integrations where supported.
Inputs	mobile GPS/baro, external device, network telemetry
Outputs	live position, track line, battery, last known location, retrieve route
Primary APIs	WS /v1/tracking/session/{id}; POST /v1/tracking/point
Acceptance criteria	Works with poor network using local buffer; privacy controls are clear and default to private.

F-016 - Accident Prevention Alerts

Field	Technical requirement
Priority	P1
Owner service(s)	alert-engine, risk-engine
Product behavior	Push/watch/audio alerts for worsening gusts, storm growth, airspace, cloudbase collapse, strong valley wind, or hard NO-GO

	transition for favorites.
Inputs	risk events, radar/lightning, station deltas, forecast updates, live pilot position
Outputs	push notification, in-app alert, voice alert, school/club dashboard flag
Primary APIs	POST /v1/alerts/rules; GET /v1/alerts/events
Acceptance criteria	Critical alerts are concise; repeated alerts deduplicated; all alerts show source and action suggestion.

F-017 - School Operations Mode

Field	Technical requirement
Priority	P2
Owner service(s)	school-service, entitlement-service
Product behavior	Student/instructor management, skill-based site choice, daily weather briefing, training records, attendance, and policy-based safety gates.
Inputs	students, instructor policies, site scores, weather history
Outputs	daily training plan, student-specific go/no-go, school log
Primary APIs	GET /v1/schools/{id}/daily-plan; POST /v1/schools/{id}/students
Acceptance criteria	Instructor can override with reason; overrides are audited; beginner safety gates stricter than public Pro defaults.

F-018 - Club Safety Dashboard

Field	Technical requirement
Priority	P2
Owner service(s)	club-service, community-service
Product behavior	Club-managed sites, member reports, local safety alerts, incident notes, station health, site maintenance tasks, and event communication.
Inputs	club members, local sites, reports, incidents, station/webcam status
Outputs	club dashboard, safety bulletin, local forecast summary
Primary APIs	GET /v1/clubs/{id}/dashboard; POST /v1/clubs/{id}/bulletins
Acceptance criteria	Club admins moderate site notes; public users see only approved local warnings.

F-019 - Competition Weather Center

Field	Technical requirement
Priority	P3
Owner service(s)	competition-service, xc-engine
Product behavior	Task weather, launch window, route risk, thermal corridor, live tracking, leaderboard integration, safety broadcast, and task cancellation evidence.
Inputs	task route, competitor positions, weather/risk scores, organizer rules
Outputs	task risk report, route forecast, broadcast alerts
Primary APIs	POST /v1/competitions/{id}/tasks; GET /v1/competitions/{id}/weather-center

Acceptance criteria	Organizer can export weather briefing PDF; route risk recalculates when task changes.
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F-020 - Rescue and Drift Support

Field	Technical requirement
Priority	P3
Owner service(s)	rescue-service, tracking-service, wind-reconstruction
Product behavior	Last-known-position tools, wind reconstruction, likely glide/drift area, track replay, and rescue handoff report.
Inputs	last GPS, wind profile, terrain, pilot wing class if known, track history
Outputs	search area estimate, last known report, exportable PDF
Primary APIs	POST /v1/rescue/case; GET /v1/rescue/{case_id}/estimate
Acceptance criteria	Clearly marked as estimate; data export is privacy-controlled and auditable.

6. Weather and Data Product Requirements

The platform must support two data maturity levels: API-first MVP and raw-model/proprietary processing for scale. The MVP may use Open-Meteo/MET-style APIs for point and small-grid forecasts. The advanced product must ingest raw GRIB/NetCDF model fields for tiles, scoring, and machine learning.

Data product	MVP source	Advanced source	Storage	Update target	Used by
Global forecast point data	Open-Meteo multi-model API	ECMWF/GFS raw GRIB subsets	forecast_timeseries + forecast_cube	Hourly to 6-hourly depending source	Site forecast, AI briefing, flyability
Nordic point forecast	MET Norway Locationforecast	MET Norway GRIB where appropriate	forecast_timeseries	As allowed by API/cache headers	Norway sites, local forecasts
Altitude wind	Open-Meteo pressure/height vars if available; model APIs	ECMWF/GFS pressure levels	forecast_cube indexed by level	Each model run	Wind layers, XC, risk
Radar / rain nowcast	Provider API / national radar	Radar mosaics	nowcast_grid	5-15 minutes when available	Now layer, alerts
Lightning	Provider API	Lightning network feed	lightning_events	Near real-time	Storm warning, no-go
Terrain	Static DEM	Copernicus DEM GLO-30/GLO-90, national DEM	terrain_tiles, slope/aspect rasters	Static with refresh	Thermals, rotor, valley wind, glide
Airspace	OpenAIP and national sources	OpenAIP + national + temporary activations	airspace_polygons	Daily or source-defined	Airspace guardian
Site observations	Pilot app reports	Stations, webcams, club sensors	observations_timeseries	Real-time to hourly	Reality layer, digital twin

6.1 Forecast Cube Data Model Requirement

forecast_cube key dimensions:

- model: ecmwf_ifs | gfs | met_norway | open_meteo_blend | corrected_site_model
- run_time: model initialization timestamp
- valid_time: forecast timestamp
- level_type: surface | agl | pressure | height_msl

- level_value: 10m, 100m_agl, 850hPa, 1500m_msl, etc.
- variable: wind_u, wind_v, gust, temp, dewpoint, cloud_low, cloud_mid, cloud_high, precipitation, cape, cin, visibility, pressure, radiation_proxy
- geometry: grid cell / tile / point
- value, units, quality_flag, source_age_seconds

6.2 Data Quality Rules

- Every forecast-derived user-facing value must include model, run_time, valid_time, and update age.
- Do not mix corrected and raw values without labeling them.
- If two models disagree beyond thresholds, confidence must drop and the reason must be shown.
- Paid tiers may unlock higher-resolution models and longer forecast horizon, but critical safety warnings are visible to all users.

7. Flyability, Risk, Thermal, and XC Requirements

The first product version should use deterministic scoring with explainable weights. Machine learning corrections can adjust input features and confidence, but the initial output should remain interpretable.

7.1 Flyability Score v1 Formula

```
flyability_score = clamp(0, 100,
    0.24 * wind_speed_score(site_limits, pilot_mode)
+ 0.18 * gust_score(max_gust, gust_spread, pilot_mode)
+ 0.14 * wind_direction_score(launch_aspect, allowed_sector, crosswind_angle)
+ 0.12 * precipitation_score(rain_rate, probability)
+ 0.10 * cloudbase_score(cloudbase_msl - launch_altitude_m, pilot_mode)
+ 0.08 * visibility_score(visibility_km)
+ 0.08 * thermal_comfort_score(thermal_strength, pilot_mode)
+ 0.06 * confidence_score(model_agreement, data_age, reality_delta)
)
```

status:

- GO: score >= 75 and no hard blockers
- MAYBE: 50 <= score < 75 or confidence low
- NO-GO: score < 50 or hard blocker present

7.2 Hard Blockers

Blocker	Trigger example	Required behavior
Thunderstorm / lightning	Lightning within configured radius; high storm risk	NO-GO; prominent alert; no AI softening
Strong gust spread	Gust spread above pilot/site threshold	NO-GO or MAYBE depending level; reason shown
Wind outside launch sector	Crosswind/tailwind outside safe sector	NO-GO for launch even if regional weather looks good
Low cloudbase / terrain obscuration	Cloudbase below launch or below safe route clearance	NO-GO or route warning
Foehn / wave / rotor flags	Pressure gradient and terrain exposure thresholds	High risk; advanced users still see warning
Radar precipitation on site	Rain nowcast over launch/route	NO-GO for free-flight launch

7.3 Risk Engine v1 Feature Vector

```
risk_features = {  
    wind_speed_surface, wind_speed_500m, wind_speed_1000m,  
    wind_shear_0_1000m, gust_spread, crosswind_angle,  
    pressure_gradient_regional, terrain_exposure_index,  
    valley_axis_alignment, slope_aspect, lee_side_flag,  
    cape, cin, lifted_index_proxy, cloud_growth_rate,  
    radar_rain_rate, lightning_distance_km,  
    station_forecast_delta, webcam_visibility_flag,  
    historical_incident_density, site_difficulty  
}
```

7.4 XC Potential v1 Outputs

- Expected launch window and climb window in local time.
- Thermal strength class: weak / moderate / strong / too strong for beginner.
- Cloudbase clearance above launch and expected route cloudbase.
- Wind drift and headwind/tailwind along common XC corridors.
- Expected distance class: local soaring, 20 km, 50 km, 100 km, competition-class.
- Landing field density and retrieve difficulty warnings.

8. AI Decision and Debrief Requirements

The AI layer must be treated as an explanation and interaction layer over deterministic backend outputs. It must not be the primary safety system. All AI answers must be grounded in structured facts produced by weather, risk, site, and profile services.

8.1 AI Context Contract

AI context object:

```
{  
    "user_question": "Can I fly Vaga tomorrow?",  
    "pilot_profile": {"level": "intermediate", "wing_class": "EN-B", "risk_mode": "standard"},  
    "site": {"id": "...", "name": "...", "launch_altitude_m": 950, "allowed_wind_sector": "NW-N-NE"},  
    "forecast_summary": [... hourly records ...],  
    "scores": {"flyability": "...", "risk": "...", "xc": "..."},  
    "hard_blockers": [...],  
    "model_agreement": {...},  
    "live_reality": {...},  
    "data_freshness": {...},  
    "allowed_response_policy": "Explain, do not invent, do not override hard blockers."  
}
```

8.2 AI Response Format

Section	Required content
Decision	GO / MAYBE / NO-GO based on backend status, not freeform LLM opinion
Best window	Time range, confidence, and reason
Main risks	Top 3 negative factors from risk engine

Why not earlier/later	Timeline explanation
Alternative sites	Only from recommendation engine results
Uncertainty	Model disagreement, missing stations, stale webcams, low sample size
Safety note	Short reminder to verify conditions at launch and obey local rules

8.3 Hallucination Control Requirements

- LLM must receive only normalized facts, not raw unbounded web context.
- Every numeric value in AI output must map to a provided context field.
- If data is missing, the response says unknown; it must not estimate unless a backend estimator supplied the value.
- Use structured JSON output internally, then render to natural language.
- Log prompt, context hash, model version, and output for audit when user enables AI history.

9. Live Reality Layer and Digital Twin Requirements

The long-term competitive moat is a data flywheel. Pilots check forecasts, fly, upload tracks/reports, and the system compares predictions to reality. This improves site-specific corrections and creates a local knowledge graph.

9.1 Minimum Digital Twin Data Before Activation

Data type	Minimum for beta activation	Purpose
Forecast history	90 days with multiple model runs	Detect recurring model bias
Weather station observations	At least one trusted nearby station or manual station mapping	Validate wind/gust error
Pilot reports	30+ validated reports or club-moderated samples	Validate subjective launch reality
Track outcomes	50+ flights or local club import	Learn thermal triggers and glide outcomes
Incidents/warnings	Any available, manually moderated	Avoid unsafe optimism

9.2 Reality Trust Score

reality_signal_trust = base_source_score

- * freshness_factor
- * spatial_relevance_factor
- * reporter_reputation_factor
- * consistency_with_other_signals

Examples:

- Club-owned calibrated station at launch: high base score
- Anonymous pilot report without history: lower base score
- Webcam AI cloud estimate: medium score unless validated
- Old report: freshness factor decays quickly

9.3 Community Reporting Anti-Abuse Requirements

- Reports include timestamp, optional GPS verification, site ID, launch/landing, wind estimate, thermals, cloudbase, crowd level, and notes.
- Reports can be private, friends-only, club-only, or public depending on user choice.
- Reputation increases with verified tracks, club membership, consistency, and moderator approval.
- Suspicious reports are downweighted, not necessarily deleted, to preserve auditability.

10. Mobile/Web UX Requirements

The product must be usable during planning, at launch, in-flight, and after landing. The UX must be fast, readable outdoors, and resilient to poor network.

Screen	Primary users	Critical components	Offline need
Home / map	All	Weather layers, site icons, time slider, altitude slider, favorites	Cached base map and favorite sites
Site detail	All	Hourly scores, wind rose, launch sector, risk reasons, live reality, webcams	Last forecast + site notes
Where to fly	Weekend/XC pilots	Drive radius, time window, ranked sites, AI explanation	Optional cached favorites
XC planner	XC/competition	Route map, thermal chain, cloudbase, wind drift, landing density	Route and terrain cache
In-flight	Pilots	Airspace, wind, glide, live tracking, alerts, minimal touch UI	Required
School dashboard	Schools	Student mode, instructor plan, daily weather report	Web preferred
Club dashboard	Clubs	Reports, station status, local warnings	Web preferred
Debrief	Pilots	Track replay, forecast comparison, coaching	Server-generated but view cache useful

10.1 Map Interaction Requirements

- One-handed mobile operation for common planning actions.
- Outdoor high-contrast mode and large icons for launch status.
- Timeline scrub with visible local time and UTC/model run information.
- Altitude selector defaults to launch-relevant levels, not meteorological jargon only.
- Picker displays both raw forecast value and corrected site forecast when available.

11. Paid Tier Entitlements and Billing Product Requirements

Paid tiers should monetize decision intelligence, depth, automation, and professional operations, not hide basic safety information. The entitlement system must be server-side and auditable.

Tier	Target	Included capabilities	Technical entitlement rules
Free	Casual pilots and acquisition	3-day basic forecast, basic map, limited favorites, basic site pages, critical safety warnings	No advanced scoring history; limited alerts; no AI except sample responses; API rate low
Pro	Weekend/intermediate pilots	10-day forecast, unlimited favorites, advanced alerts, full Flyability/Safety timelines, basic site ranking	Entitlement unlocks /flyability timeline, alert rules, longer horizon
Pro+	XC/advanced pilots	Model comparison, corrected forecast where available, XC planner, thermal maps, AI briefing, route weather	Unlocks raw-model layers, XC APIs, AI credits, digital twin details
Club	Clubs	Shared observations, club sites, member dashboard, station health, local safety bulletins	Org entitlement with roles: owner/admin/member/moderator
School	Schools/tandem	Student management, instructor briefings, training logs, strict policy gates	Org entitlement with student records and audited overrides
Event	Competitions/festivals	Task weather, live tracking, organizer alerts, exportable briefings	Event-limited entitlement, task APIs, broadcast permissions
Rescue/Authority	Rescue teams	Search support, last-known position tools, wind reconstruction	Restricted access, strong audit, explicit consent/legal workflows
API	B2B/hardware/apps	Forecast, site, scoring, alert webhooks	API keys, quotas, model-level feature flags, SLA options

11.1 Entitlement Contract

```
entitlement_check(user_id, feature_key, resource_context) -> {  
  allowed: bool,  
  tier: "free|pro|pro_plus|club|school|event|rescue|api",  
  limit_remaining: int|null,  
  reason_if_denied: string|null,  
  safety_override: bool # true when free access is granted because it is safety-critical  
}
```

Feature keys examples:

- map.altitude_layers.advanced
- forecast.horizon.10d
- ai.briefing.full
- xc.route_plan
- digital_twin.corrected_forecast
- alerts.advanced_rules
- org.school.daily_plan

12. Non-Functional Requirements

Category	Requirement	Target
Performance	Favorite site forecast page loads fast after cache warm	< 1.5 seconds p95
Map tiles	Cached weather tiles render without blocking UI	< 500 ms p95 tile response in target regions
Availability	Core forecast/site services available during flying season	99.5% MVP, 99.9% mature
Offline	Favorite site data and in-flight airspace cache available offline	At least last successful sync
Data freshness	Show source age on forecasts and observations	100% of user-visible weather values
Safety	Hard blockers cannot be hidden by personalization or paid tier	100% tests pass
Privacy	Tracks are private by default	Default private; explicit sharing required
Scalability	Support regional launch with 100k users and 10k concurrent map users	Architecture target for V1
Audit	AI decisions and safety overrides are traceable	Log context hash and decision fields
Compliance	Respect data provider terms, attribution, and branding restrictions	Required before public launch

13. MVP, V1, V2, V3 Roadmap

Phase	Goal	Must ship	Do not ship yet
MVP 0.1	Validate pilot value in one region	Site DB, Open-Meteo/MET point forecast, Flyability v1, favorites, basic map, alerts, Norway/Alps pilot testing	Raw GRIB, ML, competition, rescue
MVP 0.2	Validate recommendation engine	Where-to-fly ranking, safety score, live reports, simple AI briefing, weather station links	Full route planning, digital twin production
V1 Commercial	Paid Pro/Pro+ launch	Billing/entitlements, 10-day forecast, advanced alerts, thermal layer beta, model comparison, polished mobile/web	Enterprise rescue features
V2 Intelligence	Build moat	Digital twin beta, forecast correction, IGC debrief, webcam AI, XC route planner	Full global competition platform
V3 Platform	B2B and operations	Club/school/event portals, API tier, hardware station network, partner integrations	Overexpansion without local data validation

13.1 Initial Region Recommendation

Start in one or two high-value regions where the founder can understand pilot behavior and validate outputs. Norway is a good candidate for local knowledge and MET Norway data; the Alps are a good market-density candidate but have strong competition and more complex microclimates. The platform should not claim global precision until site-level data exists.

14. Acceptance Criteria and KPIs

KPI	Definition	Why it matters
Decision usefulness	Percent of pilots who say recommendation matched actual launch reality	Validates core product promise
Forecast correction lift	MAE reduction vs raw model for launch wind/gust	Measures digital twin value
False GO rate	Cases where app says GO and verified reality is unsafe	Primary safety KPI; must be minimized
False NO-GO rate	Cases where app says NO-GO and reality was flyable	Commercial/user trust KPI but secondary to safety
Drive success rate	User drove to recommended site and flew safely/successfully	Differentiates from generic weather apps
Alert precision	Warnings that corresponded to actual weather/risk events	Prevents alert fatigue
Pro conversion	Free users converting after recommendation/alert value	Monetization KPI
Retention	Weekly active planning users during flying season	Product-market fit
Report contribution	Validated reports per active user/month	Data flywheel strength

14.1 Feature Acceptance Test Examples

Test ID	Scenario	Expected result
AT-001	Beginner checks site with strong thermals and gusty wind	NO-GO/MAYBE with beginner-specific warning even if XC score is high
AT-002	Advanced pilot checks same site	May show MAYBE/GO for XC but risk score remains visible
AT-003	Forecast says NW 14 km/h but station reports NW 27 km/h	Reality layer lowers confidence and risk increases
AT-004	AI asked: "Can I fly now?" with stale data	AI says data is stale and refuses confident recommendation
AT-005	User without Pro+ tries XC planner	Server returns entitlement denial with upgrade reason; safety warnings still available
AT-006	Lightning detected near route	All tiers receive critical NO-GO/land-soon alert if relevant

15. Source Notes and Assumptions

This PRD uses public documentation and app descriptions as reference material for competitive positioning and feasible integrations. Exact licensing, commercial terms, usage limits, API availability, and attribution requirements must be verified during implementation and procurement.

Reference	Reason used in PRD
Windy Map Forecast API documentation	Baseline for layered map visualisation, particles, isolines, and forecast map interaction.
Ventusky public app descriptions	Baseline for broad weather-layer coverage, radar/lightning, webcams, and consumer weather UX.
Open-Meteo forecast API documentation	MVP forecast API candidate and multi-model forecast source.
MET Norway Locationforecast and API terms	Regional forecast source for Norway and example of provider

	terms/usage constraints.
ECMWF Open Data	Advanced raw model candidate using GRIB2 products.
NOAA NOMADS/GFS	Advanced raw global model candidate and fallback model source.
OpenAIP API documentation	Candidate aviation/airspace dataset source.
Copernicus DEM	Candidate terrain/elevation dataset for slope, aspect, and terrain processing.
XCTrack public feature list	Competitive baseline for in-flight flight-computer features, airspace, thermal assistant, livetracking.
Paragliding Map public feature list	Competitive baseline for launch/landing scale, stations, webcams, valley wind, and site discovery.

15.1 Key Implementation Assumptions to Validate

- Commercial use and attribution rules for each weather/data source.
- Whether API-based data is sufficient for MVP map layers or raw GRIB ingestion is needed earlier.
- Availability and licensing of paragliding launch datasets by country.
- Practical quality of pilot-submitted wind and thermal reports.
- Battery/network tradeoffs for live tracking and in-flight alerts.
- Regulatory and liability framing in each launch market.

16. Next Volumes and Engineering Handoff

This PRD defines what the platform must do and why. The next documents convert these requirements into build-ready implementation assets:

Volume	Purpose	Key outputs
Volume 02 - Technical System Architecture	Service boundaries and deployment topology	Microservices, queues, data ownership, sequence flows
Volume 03 - Database Design	Persistence implementation	Tables, indexes, PostGIS geometry, Timescale hypertables, ERD
Volume 04 - API Specification	External/internal contracts	OpenAPI endpoints, payloads, error model, WebSockets
Volume 05 - Weather Engine	Forecast ingestion and processing	GRIB pipeline, model normalization, tile generation, quality flags
Volume 06 - AI/ML Architecture	Learning and recommendation systems	Model features, training loops, digital twin validation, LLM grounding
Volume 07 - Mobile/UX	Screens and interaction design	Screen specs, offline sync, in-flight UI, accessibility

Appendix A - Example API and Data Objects

A.1 Site Object

```
{
  "id": "site_vaga_001",
  "name": "Example Launch",
  "country": "NO",
  "location": {"type": "Point", "coordinates": [10.123, 61.456]},
  "launch_altitude_m": 950,
  "landing_altitude_m": 370,
  "allowed_wind_sector_deg": {"from": 300, "to": 45},
  "wind_limits_by_mode": {
    "beginner": {"min_kmh": 4, "max_kmh": 16, "max_gust_kmh": 22},
    "intermediate": {"min_kmh": 4, "max_kmh": 22, "max_gust_kmh": 30},
    "advanced": {"min_kmh": 4, "max_kmh": 28, "max_gust_kmh": 36}
  },
  "hazards": ["lee rotor in W wind", "power line near landing"],
  "difficulty": "intermediate",
  "source_quality": "club_verified",
  "last_verified_at": "2026-06-01T00:00:00Z"
}
```

A.2 Flyability Response

```
{
  "site_id": "site_vaga_001",
  "valid_time": "2026-06-05T13:00:00+02:00",
  "pilot_mode": "intermediate",
  "status": "GO",
  "flyability_score": 82,
  "safety_risk_score": 24,
  "xc_potential_score": 71,
  "confidence": 0.78,
  "hard_blockers": [],
  "positive_reasons": ["wind inside launch sector", "cloudbase 650m above launch", "moderate thermals"],
  "negative_reasons": ["gust spread increasing after 16:00", "no trusted launch station online"],
  "sources": [
    {"model": "open_meteo_blend", "run_time": "2026-06-05T06:00:00Z"},
    {"observation": "station_123", "observed_at": "2026-06-05T10:45:00Z"}
  ]
}
```

A.3 Alert Rule Object

```
{
  "id": "alert_001",
  "user_id": "user_123",

```

```
"site_ids": ["site_vaga_001"],
"rule_type": "flyability_window",
"conditions": {
  "min_flyability": 75,
  "max_risk": 35,
  "allowed_hours_local": ["10:00", "17:00"],
  "pilot_mode": "intermediate"
},
"delivery": {"push": true, "email": false},
"entitlement_required": "pro"
}
```

Final Note

The platform should be built as a safety-aware decision system from day one. The first marketable product can be small, but the data model must already anticipate site-specific correction, digital twins, user skill modes, and paid entitlements. That is the difference between another weather map and a defensible paragliding intelligence platform.